Problem Set 6 Solutions

QTM 220

Due 1 December at 2:00pm ET on Canvas. Questions about submission requirements should be directed to your TA.

California CPS Revisited

(a) Using the California sample of 25-35 year olds, regress income on education and age (in an additive model). Report your results in a table, including the standard errors that R automatically calculates.

```r
cps<- read.csv("California_CPS_degrees.csv")

mod1<- lm(income~ education + age,data=cps)

summary(mod1)
```

```
Call:
lm(formula = income ~ education + age, data = cps)

Residuals:
    Min      1Q  Median      3Q     Max
-226476  -62493  -20343   36193  752685

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -137806.3    24024.5  -5.736  1.1e-08 ***
education     16479.0      889.8  18.520  < 2e-16 ***
age            1051.6      694.4   1.514     0.13
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 101800 on 2243 degrees of freedom
Multiple R-squared:  0.1345,	Adjusted R-squared:  0.1337 
F-statistic: 174.2 on 2 and 2243 DF,  p-value: < 2.2e-16
```

(b) Create a matrix of your X information for your model (education and age) and include a column of 1s to estimate an intercept. Using the following matrix functions in base R, calculate $(X'X)^{-1}X'Y$ and confirm your resulting $\hat{\beta}s$ are the same as your lm results:

Let X be your matrix in R. Use $\% * \%$ to multiply matrices, i.e. $X\% * \%X$ will yield XX

Use $t(X)$ to take the transpose of matrix X

Use solve(X) to take the inverse of matrix X

```r
df.mat<- data.frame(constant=rep(1,length(cps$income)), edu=cps$education, age=cps$age)

X<- data.matrix(df.mat)

solve(t(X)%*%X)%*%t(X)%*%(cps$income)
```

```
                [,1]
constant -137806.340
edu        16478.997
age         1051.582
```

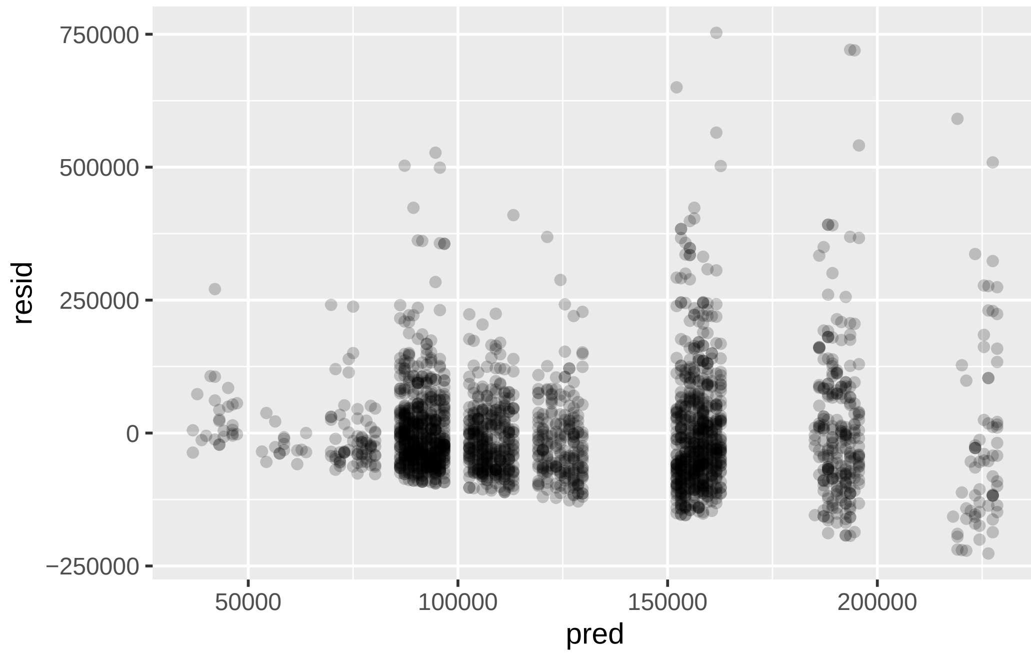
(c) Extract the predicted values and residuals from your model using predict(model name) and residuals(model name), then create a scatter plot of your predicted values and residuals. What do you see? Would you guess the errors are homoskedastic or heteroskedastic? Why?

```r
pred<- predict(mod1)
resid<- residuals(mod1)

df<- data.frame(pred=pred,resid=resid)


rplot<- ggplot(df, aes(x=pred,y=resid)) + geom_point(alpha=.2)

rplot
```



(d) install the sandwich package and get heteroskdasticity consistent standard errors for your model

```
sandwich(mod1)
```

```
              (Intercept)      education            age
(Intercept)     666941234 -15029237.613 -15481152.468
education       -15029238   1118548.385      2254.142
age             -15481152      2254.142    517429.329
```

```
sqrt(diag(sandwich(mod1)))
```

```
(Intercept)   education         age
 25825.2054   1057.6145    719.3256
```

(e) do matrix algebra to get the same thing

```
D<- diag(resid^2)

dim(D)
```

```
[1] 2246 2246
```

```
betavar<- solve(t(X)%*%X)%*%t(X)%*%D%*%X%*%solve(t(X)%*%X)

sqrt(diag(betavar))
```

```
  constant         edu         age 
25825.2054   1057.6145    719.3256 
```

(f) Bootstrap!

(g) Compare the standard errors you got from R in part (a) to the robust (sandwich) you calculated in parts (e) and (f). Why are they different? How do these compare to what you estimated from the bootstrap?

Causal Warm Up: Bone Density

Doctors (and researchers) hypothesize that higher calcium intake causes greater bone density (stronger bones). To test this, a clinic runs an experiment where patients are treated with a calcium supplement or a placebo for 1 year, after which their bone density is measured. Patients are first classified as old or young, then treatment assignment is randomized within the groups. Bone Density T-Scores indicate how "normal" someone's bone density is—0 is typical, positive numbers are better, negative numbers are worse (a threshold score of -2.5 is used to diagnose osteoporosis).

The resulting data is below.

Age Classification	Calcium Supplement	Bone Density T-Score
Old	1	-2
Old	0	-3
Old	1	-2
Old	0	0
Old	1	-1
Old	0	0
Old	0	-2
Old	0	0
Young	0	0
Young	1	1
Young	0	-1
Young	1	1.5

(a) What is the probability of receiving treatment for the young? What is the probability of receiving treatment for the old?

Old: $\frac{3}{8}$; Young: $\frac{1}{2}$

(b) Using simple difference in means, what is the treatment effect of the calcium supplement on bone density among the young? What is the treatment effect of the calcium supplement among the old?

(c) What is the average of these treatment effects, weighting by the number of participants in each category?

(d) If you ignored the blocked randomization and calculated the ATE using the difference in means in treated and control, what would your estimate be?

(e) Input the data in R, creating a binary variable for age classification. Run two regressions: a simple additive of treatment on bone density score, and an interactive with age, treatment, and age*treatment. Write the equations for your two models using βs. Report your results in a table. Which model corresponds with which estimate of the treatment effect? Fill in the table below with the βs notation from your equations and the estimates from your model.

	βs
Avg Bone Density [Treated, Old]	
Avg Bone Density [Control, Old]	
Avg Bone Density [Treated, Young]	
Avg Bone Density [Control, Young]	
$\hat{ATE}_{Old}$	
$\hat{ATE}_{Young}$	

Neto and Cox

Use netocox.csv for Neto and Cox's 1997 data on the effect of runoff elections on the number of effective presidential candidates. The following variables are included:

- RUNOFF : whether the country has a runoff electoral system (treatment)
- ENPRES : the effective number of presidential candidates (outcome)
- ENETH : the effective number of ethnic groups

For the purposes of this exercise, let's assume that runoff systems are randomly assigned such that we can interpret estimates causally.

(a) Are the number of effective ethnic groups balanced across treatment and control countries? Make a (small) balance table reporting the means of the pre-treatment covariate.

(b) Fit an additive linear model of runoff and ethnic groups on number of presidential candidates. Calculate the Average Partial Derivative ($A\hat{P}D$) of this model.

```
mod2<- lm(netocox$ENPRES~netocox$RUNOFF+netocox$ENETH)
summary(mod2)
```

```
Call:
lm(formula = netocox$ENPRES ~ netocox$RUNOFF + netocox$ENETH)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.3094 -1.0066 -0.2142  0.9920  1.9826 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)  
(Intercept)      2.2629     0.8691   2.604   0.0219 *
netocox$RUNOFF   0.6311     0.6087   1.037   0.3187  
netocox$ENETH    0.3661     0.4986   0.734   0.4759  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.211 on 13 degrees of freedom
Multiple R-squared:  0.1206,	Adjusted R-squared:  -0.01464 
F-statistic: 0.8918 on 2 and 13 DF,  p-value: 0.4336
```

(c) Fit an interactive linear model of runoff and ethnic groups on number of presidential candidates. Calculate the Average Partial Derivative ($A\hat{P}D$) of this model.

(d) Using the interactive model, impute $Y(1)$s for the no runoff countries and $Y(0)$s for the runoff countries. Fill in your estimates in the table below, then calculate the ATE of runoff on presidential candidates.

Country	Runoff?	Y(0)	Y(1)
Argentina	0		
Austria	1		
Brazil	1		
Colombia	0		
Costa Rica	1		
Dominican Republic	0		
El Salvador	1		
Ecuador	1		
Finland	0		
France	1		
Iceland	0		
Korea	0		
Peru	1		
Portugal	1		
USA	0		
Venezuela	0		

Other?

Menopausal Hormone Therapy

compare RCT and observational studies

read some stuff

reflect